

Marginal Distribution

45

We have noted earlier that if the joint probability function $f(x, y)$ of two random variables X and Y is known, the probability functions of X and Y can be ~~def~~ derived separately.

In the case when the distribution of X is derived from $f(x, y)$, the resulting distribution is called marginal distribution of X .

Similarly the probability density of Y derived from the joint distribution of X and Y is known as the marginal distribution of Y .

Here, we will denote these two distributions by $g(x)$ and $h(y)$, respectively, when they are derived from joint distribution $f(x, y)$.

Thus, given the joint probability $f(x, y)$ of two discrete random variables X and Y , the probability distribution of X alone is

$$g(x) = \sum_y f(x, y)$$

and that of Y alone is

$$h(y) = \sum_x f(x, y)$$

These two are the marginal distributions of two discrete random variables X and Y , respectively.

Now, when X and Y are continuous,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy; \text{ for } -\infty < x < \infty$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx; \text{ for } -\infty < y < \infty$$

Marginal distributions are indeed probability distributions, since they satisfy all the properties of a probability distribution.

For continuous case,

$$\int_{-\infty}^{\infty} g(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\begin{aligned} \text{and } P[a < X < b] &= P[a < X < b, -\infty < Y < \infty] \\ &= \int_a^b \int_{-\infty}^{\infty} f(x, y) dx dy = \int_a^b g(x) dx \end{aligned}$$

Defⁿ :

Let, $F(x, y)$ be the joint CDF of (X, Y) . Then the marginal CDF of X is

$$F_1(x) = \lim_{y \rightarrow \infty} F(x, y).$$

Similarly, the marginal CDF of Y is

$$F_2(y) = \lim_{x \rightarrow \infty} F(x, y).$$

The probability function or PDF of X associated with the marginal CDF of X is called the marginal P.F or marginal PDF of X .

The marginal probability function and marginal PDF can be obtained from joint probability function and joint PDF, respectively.

Example: Suppose X and Y have the following joint probability distribution

	X				
Y	0	1	2	3	Row Sum
0	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{4}{8}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0	$\frac{4}{8}$
Column Sum	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

Find the marginal distributions of X and Y .

Sol: For the random variable X

$$g(0) = P(X=0) = \sum_{y=0}^1 f(0,y) = f(0,0) + f(0,1) \\ = 0 + \frac{1}{8} = \frac{1}{8}$$

$$g(1) = P(X=1) = \sum_{y=0}^1 f(1,y) = f(1,0) + f(1,1) \\ = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

$$g(2) = P(X=2) = \sum_{y=0}^1 f(2,y) = f(2,0) + f(2,1) \\ = \frac{2}{8} + \frac{1}{8} = \frac{3}{8}$$

$$g(3) = P(X=3) = \sum_{y=0}^1 f(3,y) = f(3,0) + f(3,1) = \frac{1}{8} + 0 = \frac{1}{8}$$

Similarly for Y

$$h(0) = P(Y=0) = \sum_{x=0}^3 f(x,0) = f(0,0) + f(1,0) + f(2,0) + f(3,0)$$

$$= 0 + \frac{1}{8} + \frac{2}{8} + \frac{1}{8} = \frac{4}{8}$$

$$h(1) = P(Y=1) = \sum_{x=0}^3 f(x,1) = f(0,1) + f(1,1) + f(2,1) + f(3,1)$$

$$= \frac{1}{8} + \frac{2}{8} + \frac{1}{8} + 0 = \frac{4}{8}$$

The marginal distributions derived above may also be represented as follows:

Marginal distribution of X :

x	0	1	2	3	Sum
$f(x)$	$\frac{4}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

Marginal distribution of Y :

y	0	1	Sum
$f(y)$	$\frac{4}{8}$	$\frac{4}{8}$	1

The above two columns show that marginal distributions are necessarily probability distributions.

Example : Find the marginal densities of X and Y from the following joint density function and verify that marginal distributions are also probability distributions.

$$f(x, y) = \frac{1}{8}(6 - x - y), \text{ for } 0 < x < 2, 2 < y < 4$$

$$= 0, \text{ otherwise}$$

Also compute $P(X + Y < 3)$ and $P(X < 1.5, Y < 2.5)$

Solⁿ : By definition, the marginal density of X is

$$g(x) = \frac{1}{8} \int_2^4 f(x, y) dy$$

$$= \frac{1}{8} \int_2^4 (6 - x - y) dy$$

$$= \frac{1}{8} \left[6y - xy - \frac{y^2}{2} \right]_2^4$$

$$= \frac{1}{8} \left[(24 - 4x - 8) - (12 - 2x - 2) \right]$$

$$= \frac{1}{8} (16 - 4x - 10 + 2x)$$

$$= \frac{1}{4} (3 - x); \text{ for } 0 < x < 2$$

Similarly, the marginal density of Y is

$$\begin{aligned}h(y) &= \frac{1}{8} \int_0^2 f(x, y) dx \\&= \frac{1}{8} \int_0^2 (6 - x - y) dx \\&= \frac{1}{8} \left[6x - \frac{x^2}{2} - \frac{xy}{1} \right]_0^2 = \frac{1}{8} [12 - 2y - 2y]\end{aligned}$$

$$\Rightarrow h(y) = \frac{1}{4} (5 - y) \quad , \quad \text{for } 2 < y < 4$$

Now, we need to verify that $g(x)$ and $h(y)$ are probability distributions.

It is clear that in the given range of the variables X and Y ; $g(x) \geq 0$ & $h(y) \geq 0$

$$\text{Also, } \int_0^2 g(x) dx = \frac{1}{4} \int_0^2 (3 - x) dx = \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_0^2 = 1$$

$$\& \int_2^4 h(y) dy = \frac{1}{4} \int_2^4 (5 - y) dy = \frac{1}{4} \left[5y - \frac{y^2}{2} \right]_2^4 = 1$$

The $g(x)$ & $h(y)$ thus satisfy all the conditions of a density function.

Now,

$$P(X+Y < 2) = \frac{1}{8} \int_{x=0}^2 \int_{y=2-x}^{3-x} (6-x-y) dx dy$$

$$= \frac{1}{8} \int_{x=0}^2 \left[\int_{y=2-x}^{3-x} (6-x-y) dy \right] dx$$

$$= \frac{1}{8} \int_{x=0}^2 \left[6y - xy - \frac{y^2}{2} \right]_{2-x}^{3-x} dx$$

$$= \frac{1}{8} \int_{x=0}^2 \left[\left\{ 6(3-x) - x(3-x) - \frac{(3-x)^2}{2} \right\} - \left\{ 12 - 2x - 2 \right\} \right] dx$$

$$= \frac{1}{8} \int_{x=0}^2 \left(18 - 6x - 3x + x^2 - \frac{9 - 6x + x^2}{2} - 10 + 2x \right) dx$$

$$= \frac{1}{8} \int_0^2 \left(8 - 4x + x^2 - \frac{9}{2} + 3x - \frac{x^2}{2} \right) dx$$

$$= \frac{1}{8} \int_0^2 \left(\frac{x^2}{2} - 4x + \frac{7}{2} \right) dx = \frac{1}{24}$$

and,

$$P(X < 3/2, Y < 5/2) = \frac{1}{8} \int_{x=0}^{3/2} \int_{y=0}^{5/2} (6-x-y) dy dx$$

$$= \frac{1}{8} \int_{x=0}^{3/2} \left[6y - xy - \frac{y^2}{2} \right]_0^{5/2} dx$$

$$= \frac{1}{8} \int_{x=0}^{3/2} \left(\frac{15}{2} - \frac{x}{2} \right) dx$$

$$= \frac{1}{8} \left[\frac{15x}{2} - \frac{x^2}{4} \right]_0^{3/2}$$

$$= \frac{1}{8} \left(\frac{45}{4} - \frac{9}{16} \right)$$

$$= \frac{1}{8} \times \frac{36}{16} = \frac{9}{32}$$

0

Example : The joint probability function of a discrete bivariate random variable is as follow

$$f(x, y) = \begin{cases} \frac{1}{36} & \text{for } 1 \leq x = y \leq 6 \\ \frac{2}{36} & \text{for } 1 \leq y < x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal probability functions of X & Y

Solⁿ : The marginal probability function of X is

$$g(x) = \sum_y f(x, y)$$

$$= \sum_{y=1}^6 f(x, y)$$

$$= f(x, x) + \sum_{y < x}^6 f(x, y) + \sum_{y > x}^6 f(x, y)$$

$$= \frac{1}{36} + (x-1) \frac{2}{36} + 0$$

$$= \frac{2x-1}{36}$$

Thus,
$$g(x) = \begin{cases} \frac{2x-1}{36} ; & \text{for } x=1, 2, 3, 4, 5, 6 \\ 0 ; & \text{otherwise} \end{cases}$$

Marginal ~~distribution~~ probability function
of Y is

$$\begin{aligned}h(y) &= \sum_x f(x, y) \\&= \sum_{x=1}^6 f(x, y) \\&= f(y, y) + \sum_{x < y} f(x, y) + \sum_{x > y} f(x, y) \\&= \frac{1}{36} + 0 + (6-y) \frac{2}{36}\end{aligned}$$

$$= \frac{13-2y}{36}$$

Thus,

$$h(y) = \begin{cases} \frac{13-2y}{36}; & \text{for } y = 1, 2, 3, 4, 5, 6 \\ 0; & \text{elsewhere} \end{cases}$$

Bivariate cumulative distribution functions:

The joint distribution function or joint cumulative distribution function (joint CDF) F of two dimensional random variable (X, Y) is defined as a function

$$F(x, y) = P(X \leq x \text{ and } Y \leq y); -\infty < x < \infty, \text{ and } -\infty < y < \infty$$

- $F(x, y)$ is a monotonic increasing function in x for each fixed y and is a monotonic increasing function in y for each fixed x .

Here it is noted that the CDF of random variable X is $F_1(x) = \lim_{y \rightarrow \infty} F(x, y)$

- The CDF of random variable Y is $F_2(y) = \lim_{x \rightarrow \infty} F(x, y)$

□ If the bivariate random variable (X, Y) is discrete, then

$$F(x, y) = P(X \leq x \text{ and } Y \leq y)$$

$$= \sum_{r \leq x} \sum_{s \leq y} F(r, s) \text{ for } -\infty < x < \infty \text{ and } -\infty < y < \infty$$

■ If the bivariate random variable (X, Y) is continuous then

$$F(x, y) = P(X \leq x \text{ and } Y \leq y)$$

$$= \int_{-\infty}^y \int_{-\infty}^x f(r, s) dr ds$$

for $-\infty < x < \infty$ and $-\infty < y < \infty$

■ If the bivariate random variable (X, Y) is continuous, then

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

Note:

For given numbers $a < b$ and $c < d$

$$P(a \leq X \leq b \text{ and } c \leq Y \leq d)$$

$$= P(a \leq X \leq b \text{ and } Y \leq d) - P(a \leq X \leq b \text{ and } Y \leq c)$$

$$= P(X \leq b \text{ and } Y \leq d) - P(X \leq a \text{ and } Y \leq d)$$

$$- P(X \leq b \text{ and } Y \leq c) + P(X \leq a \text{ and } Y \leq c)$$

$$= F(b, d) - F(a, d) - F(b, c) + F(a, c)$$



Example: The joint PDF of (X, Y) is

$$f(x, y) = \begin{cases} \frac{1}{8}(x+y); & \text{for } 0 < x < 2 \text{ and } 0 < y < 2 \\ 0; & \text{otherwise} \end{cases}$$

Find its joint CDF.

Solⁿ: The joint CDF of (X, Y) is as follows:

For $x < 0$ or $y < 0$, $F(x, y) = 0$

For $0 \leq x < 2$ or $0 \leq y < 2$

$$F(x, y) = \int_0^x \int_0^y \frac{1}{8}(x+y) dy dx$$

$$= \frac{1}{8} \int_0^x \left[xy + \frac{y^2}{2} \right] dx$$

$$= \frac{1}{8} \left[\frac{x^2}{2} y + x \frac{y^2}{2} \right]_0^x$$

$$\Rightarrow F(x, y) = \frac{1}{16} xy(x+y)$$

For $0 \leq x < 2$ and $y \geq 2$

$$F(x, y) = \int_0^x \int_0^2 \frac{1}{8}(x+y) dy dx$$

$$\begin{aligned}
&= \frac{1}{8} \int_0^2 \left[xy + \frac{y^2}{2} \right]_0^2 dx \\
&= \frac{1}{4} \int_0^2 (x+1) dx \\
&= \frac{1}{4} \left[\frac{x^2}{2} + x \right]_0^2 \\
&= \frac{1}{8} (x^2 + 2x) = \frac{1}{8} x(x+2)
\end{aligned}$$

For $x \geq 2$ and $0 \leq y < 2$

$$\begin{aligned}
F(x, y) &= \int_0^2 \int_0^y \frac{1}{8} (x+y) dy dx \\
&= \frac{1}{8} \int_0^2 \left[xy + \frac{y^2}{2} \right]_0^y dx \\
&= \frac{1}{8} \int_0^2 \left[xy + \frac{y^2}{2} \right] dx \\
&= \frac{1}{8} \left[\frac{x^2}{2} y + \frac{xy^2}{2} \right]_0^2 \\
&= \frac{1}{16} [4y + 2y^2]
\end{aligned}$$

$$\therefore F(x, y) = \frac{1}{8} y(y+2)$$

For $x \geq 2$ and $y \geq 2$

$$F(x, y) = 1$$

Thus the complete joint CDF of (X, Y) is given by

$$F(x, y) = \begin{cases} 0; & \text{for } x < 0 \text{ or } y < 0 \\ \frac{1}{16} xy(x+y); & \text{for } 0 \leq x < 2 \text{ and } 0 \leq y < 2 \\ \frac{1}{8} x(x+2); & \text{for } 0 \leq x < 2 \text{ and } y \geq 2 \\ \frac{1}{8} y(y+2); & \text{for } x \geq 2 \text{ and } 0 \leq y < 2 \\ 1; & \text{for } x \geq 2 \text{ and } y \geq 2 \end{cases}$$

Example : The joint CDF for water and electric demand (X, Y) is

$$F(x, y) = \begin{cases} 0; & \text{for } x < 4 \text{ or } y < 2 \\ \frac{(x-4)(y-2)}{19208}; & \text{for } 4 \leq x < 200 \\ & \text{and } 2 \leq y < 100 \\ \frac{x-4}{196}; & \text{for } 4 \leq x < 200 \text{ and } y \geq 100 \\ \frac{y-2}{98}; & \text{for } x \geq 200 \text{ and } 2 \leq y < 100 \\ 1; & \text{for } x \geq 200 \text{ and } y \geq 100. \end{cases}$$

(a) Find the joint PDF of (X, Y) ;

(b) $P(50 \leq X \leq 100, 50 \leq Y \leq 75)$;

(c) The CDF of X ;

(d) The CDF of Y ;

Solⁿ: (a) For $4 \leq x \leq 200$ and $2 \leq y \leq 100$

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

$$= \frac{\partial}{\partial x} \left(\frac{(x-4)(y-2)}{19208} \right)$$

$$= \frac{\partial}{\partial x} \left\{ \frac{(x-4)}{19208} \right\}$$

$$= \frac{1}{19208}$$

Thus, $f(x, y) = \begin{cases} \frac{1}{19208} & ; \text{ for } 4 \leq x \leq 200 \\ & \text{ and } 2 \leq y \leq 100 \\ 0, & \text{ otherwise} \end{cases}$

Ans

$$(b) P(50 \leq X \leq 100, 50 \leq Y \leq 75)$$

$$= F(100, 75) - F(50, 75) - F(100, 50) + F(50, 50)$$

$$= \left\{ \frac{(100-4)(75-2)}{19208} \right\} - \left\{ \frac{(50-4)(75-2)}{19208} \right\} \\ - \left\{ \frac{(100-4)(50-2)}{19208} \right\} + \left\{ \frac{(50-4)(50-2)}{19208} \right\}$$

$$= \frac{625}{9604} \quad \underline{\text{Ans}}$$

(c) If $4 \leq x < 200$ and $y \geq 100$, then $F(x, y) = F(x, 100)$, and it follows that

$$F(x, y) = \frac{x-4}{196}$$

Thus by letting $y \rightarrow \infty$, the CDF of X is

$$F_1(x) = \begin{cases} 0; & \text{for } x < 4 \\ \frac{x-4}{196}; & \text{for } 4 \leq x < 200 \\ 1; & \text{for } x \geq 200 \end{cases}$$

(d) If $2 \leq y < 100$ and $x \geq 200$, then

$F(x, y) = F(200, y)$ and it follows that

$$F(x, y) = \frac{y-2}{98}$$

Thus letting $x \rightarrow \infty$, the CDF of Y is

$$F_2(y) = \begin{cases} 0; & \text{for } y < 2, \\ \frac{y-2}{98}; & \text{for } 2 \leq y < 100, \\ 1; & \text{for } y \geq 100 \end{cases}$$